

(a) ASCII map

[illegible]

HP: 16/16 AC: 6 Dlvl: 3 Turn: 241 XP: 1 \$: 3 Pos: (39,14)

(b) JSON entities

```
{
  "player": [39, 14],
  "entities": [
    {"kind": "item", "x": 32, "y": 3, "desc": "tool"},
    {"kind": "item", "x": 40, "y": 13, "desc": "potion"},
    {"kind": "monster", "x": 57, "y": 13, "desc": "bat"},
    {"kind": "monster", "x": 39, "y": 14, "desc": "valkyrie"},
    {"kind": "item", "x": 31, "y": 16, "desc": "door (closed)"},
    {"kind": "item", "x": 40, "y": 16, "desc": "potion"},
    {"kind": "item", "x": 41, "y": 16, "desc": "food/corps"},
    {"kind": "item", "x": 42, "y": 16, "desc": "armor"},
    {"kind": "monster", "x": 33, "y": 17, "desc": "grid bug"},
    {"kind": "monster", "x": 53, "y": 17, "desc": "brown mold"},
    {"kind": "door", "x": 71, "y": 1, "desc": "door (closed)"},
    {"kind": "door", "x": 61, "y": 3, "desc": "door (closed)"},
    {"kind": "stair", "x": 32, "y": 4, "desc": "stairs UP"},
    {"kind": "door", "x": 72, "y": 6, "desc": "door (closed)"},
    {"kind": "door", "x": 74, "y": 6, "desc": "door (closed)"},
    {"kind": "door", "x": 5, "y": 9, "desc": "door (closed)"},
    {"kind": "stair", "x": 42, "y": 14, "desc": "stairs DOWN"},
    {"kind": "door", "x": 31, "y": 16, "desc": "door (closed)"},
    {"kind": "door", "x": 63, "y": 16, "desc": "door (closed)"},
    {"kind": "door", "x": 63, "y": 18, "desc": "door (closed)"},
    {"kind": "door", "x": 11, "y": 19, "desc": "door (closed)"}
  ]
}
```

(c) Lists, map hidden

```
=== MAP (hidden; call reveal(x1,y1,x2,y2) to view a region) ===
```

=== STATUS ===

HP: 16/16 AC: 6 Dlvl: 3 Turn: 241 XP: 1 \$: 3 Pos: (39,14)
Character: valkyrie (human, neutral)

=== VISIBLE FEATURES ===

stairs DOWN at (42,14); stairs UP at (32,4); door (open/gap) at (70,2), (55,3), (22,4), (36,7), (6,8), (43,13), (51,13), (56,13), (8,15), (27,15), (56,15), (66,15), (71,15), (12,16), (43,16), (51,16), (56,17), (34,18); door (closed) at (71,1), (61,3), (72,6), (74,6), (5,9), (31,16), (63,16), (63,18), (11,19); food/corpse at (41,6); potion at (40,13), (40,16); armor at (42,16); weapon at (32,3)

=== VISIBLE MONSTERS ===

grid bug at (33,17), 6 steps SW; brown mold at (53,17), 14 steps E;
bat at (57,13), 18 steps E

=== MESSAGES ===

There is a doorway here.